Online Chatter Predicting the Stock Market

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Does online chatter predict what a stock does next?

Six sources of public discussion — mainstream news, Hacker News, five subreddits, investor posts and SEC filings — scored for sentiment across eleven years and matched against daily prices for roughly two thousand companies.

The short answer: yes, but modestly — and the edge has been shrinking.

Signals analysed	Positive buzz accuracy	Negative buzz accuracy	Best strategy vs S&P 500
39,171	53.8%	45.7%	3.2×
	↑ +3.8 pts vs chance	↓ -4.3 pts vs chance	↑ 141,694 vs 43,868

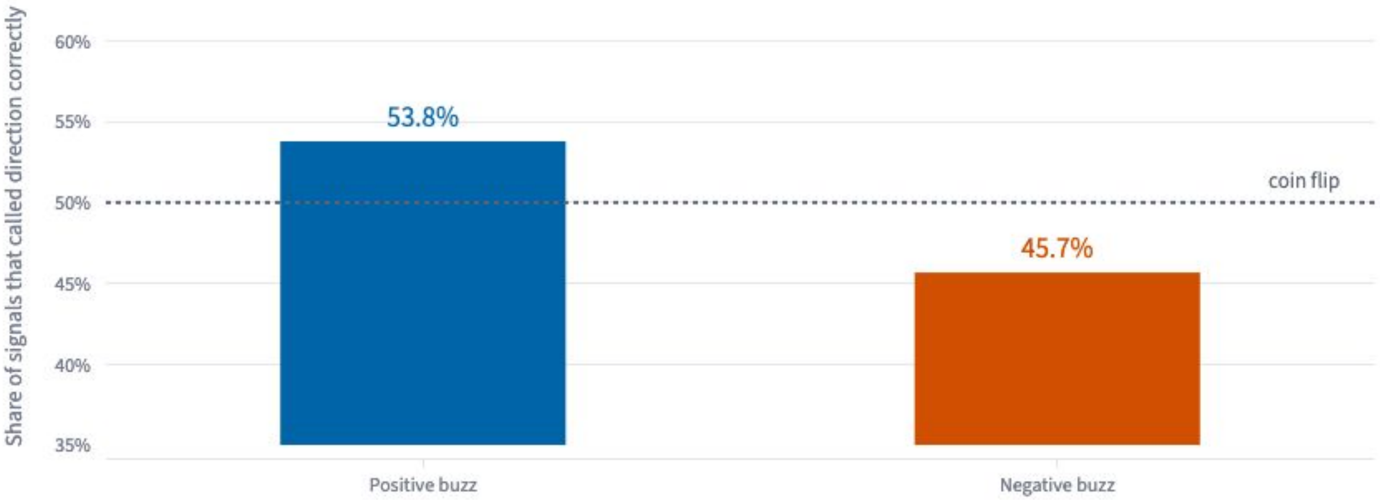
1 · Positive sentiment carries a small but real edge

When public discussion turns positive, prices rose over the following 5 trading days **53.8%** of the time — 3.8 points better than chance, across 30,097 signals. The margin is small by design: an obvious edge would already have been competed away. It is consistent enough to build a strategy on, which is what finding 3 tests.

› What about negative sentiment?

Does positive buzz beat chance? Yes, by 3.8 points

Judged over the following 5 trading days. Negative signals are shown for completeness on a much smaller sample (9,074 against 30,097) and are discussed ab

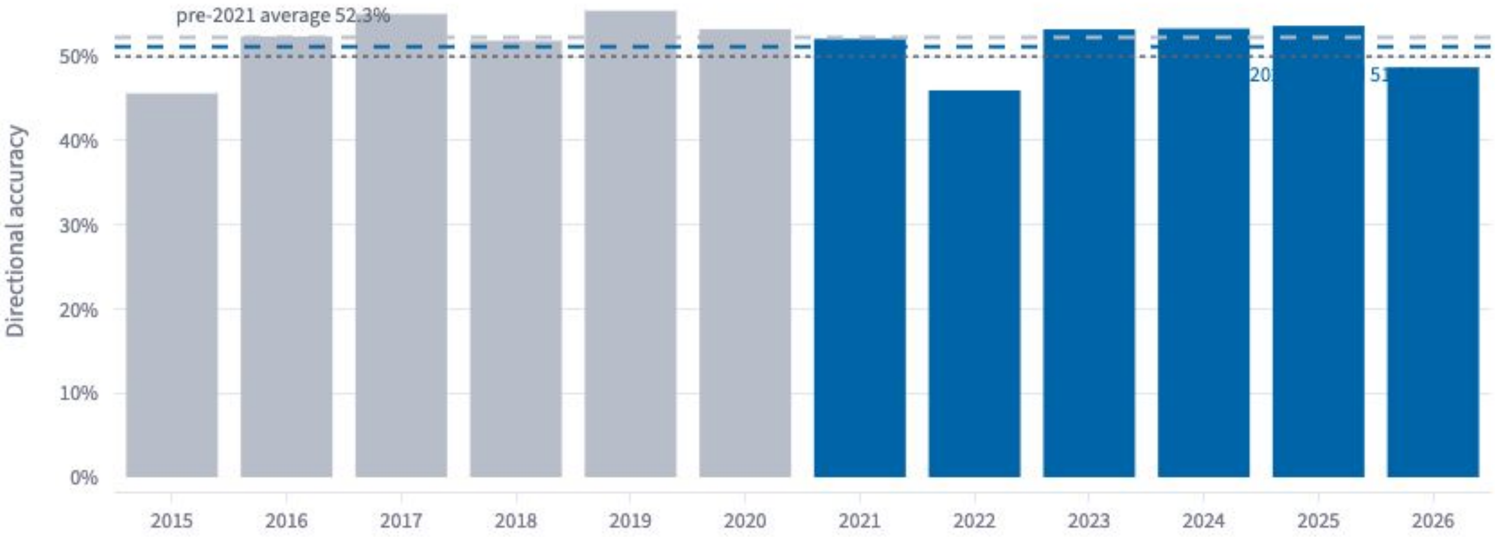


2 · The edge has faded since the retail trading boom

Accuracy averaged **52.3%** before 2021 and **51.2%** afterwards. Commission-free trading, a surge in retail participation and funds mining social data directly all arrived in that window — consistent with an edge being competed away as more people traded on the same public signals.

Has the edge faded? Accuracy fell 1.1 points after 2021

Annual hit rate. Years with fewer than 100 signals are excluded



3 · It beats the market, but not comfortably

A buzz-driven sector rotation turned $10, 000into * *$141,694** against **$43,868** for the index, after transaction costs. The catch is in the lower panel: it trailed the index in **44%** of rolling twelve-month windows. The gains arrive in bursts separated by long stretches of underperformance.

Does it survive execution? Sector Rotation finished ahead of both benchmarks

Starting from $10,000 in 2015, including transaction costs

Strategies

S&P 500 (SPY) — passive benchmark, never touched. **Buy & Hold** — equal weight across all tickers.

Sector Rotation — weekly rotation into the sector with the strongest buzz. **Position Trader** — 21-day holds on high-conviction signals.

Which strategy came out ahead? Sector Rotation

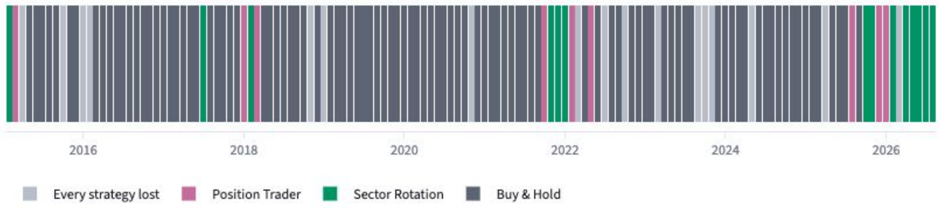
Best strategy finished at $141,694, about 3.2× the index

S&P 500 (SPY)	Buy & Hold	Sector Rotation	Position Trader
$43,868	$72,005	$141,694	$85,350
↑ +338.9%	↑ +622.0%	↑ +1316.9%	↑ +753.5%

Monthly leader

Which strategy was ahead in any given month?

Grey marks months where every strategy lost money

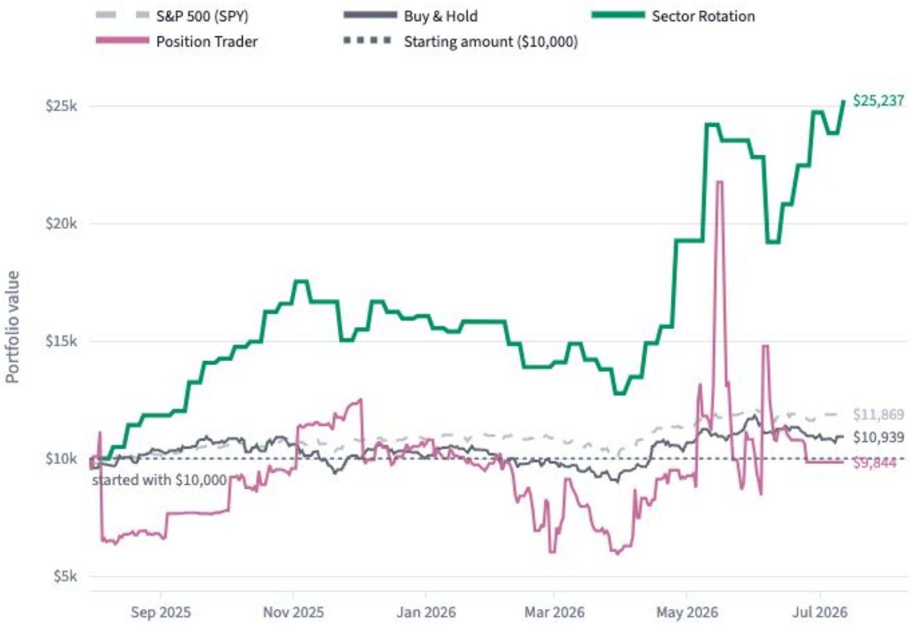


What If

Put in an amount and see how each strategy would have handled it over the date range set in the sidebar.

What would $10,000 have become? $25,237 in Sector Rotation

From July 2025 to July 2026, including transaction costs



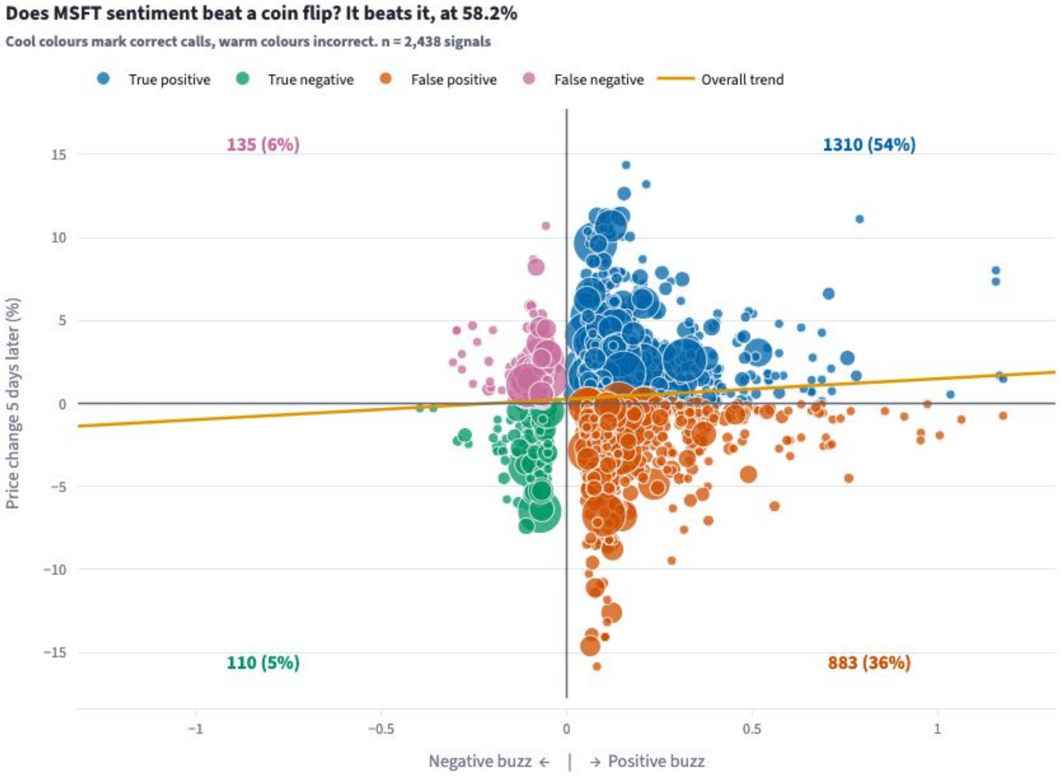
S&P 500 (SPY)	Buy & Hold	Sector Rotation	Position Trader
$11,869	$10,939	$25,237	$9,844
↑ +$1,869 (+18.7%)	↑ +$939 (+9.4%)	↑ +$15,237 (+152.…	↑ −$156 (-1.6%)

How many trades actually made money? 12 of 18, a 67% win rate

Average return -0.47% per trade

Signal Quality

Every dot is one day where sentiment crossed the neutral threshold, placed by how strong the signal was and what the price did next. Dot size reflects how many stories drove that reading.



Accuracy	True positive	False positive
58.2%	1310 ↑ 54%	883 ↑ 36%

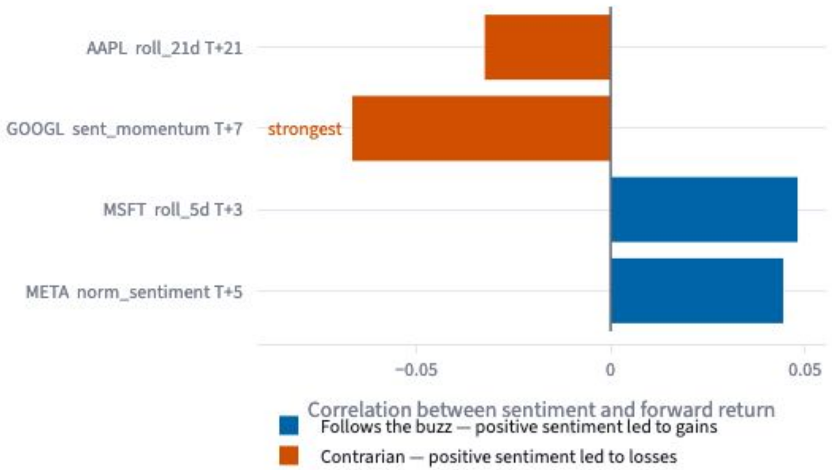
	True negative	False negative
	110 ↑ 5%	135 ↑ 6%

Correlation

How strongly does sentiment lead price for each company? Bars to the right mean positive buzz preceded gains. Bars to the left are contrarian — there, negative buzz was the more useful signal.

Which company has the most predictive chatter? GOOGL

4 of 4 tickers in this sector clear the 200-observation minimum



Has sentiment toward GOOGL shifted over time? It has cooled

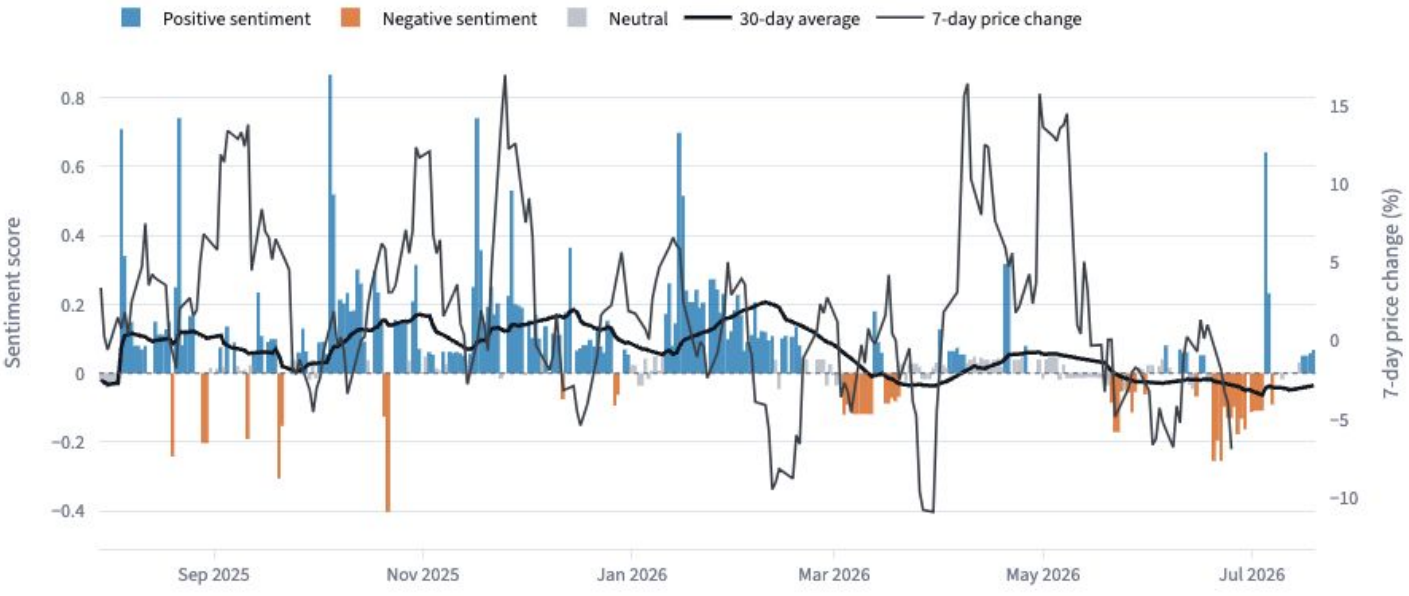
Rolling 90-day share of active sentiment days that were positive. Shading marks stretches where negati

Sentiment vs Price

Sentiment scored from six sources, shown against how the price actually moved. Blue bars are positive days, orange negative, grey inside the neutral threshold.

How did sentiment toward GOOGL behave? It leaned positive

342 days carried a sentiment reading | average score +0.063

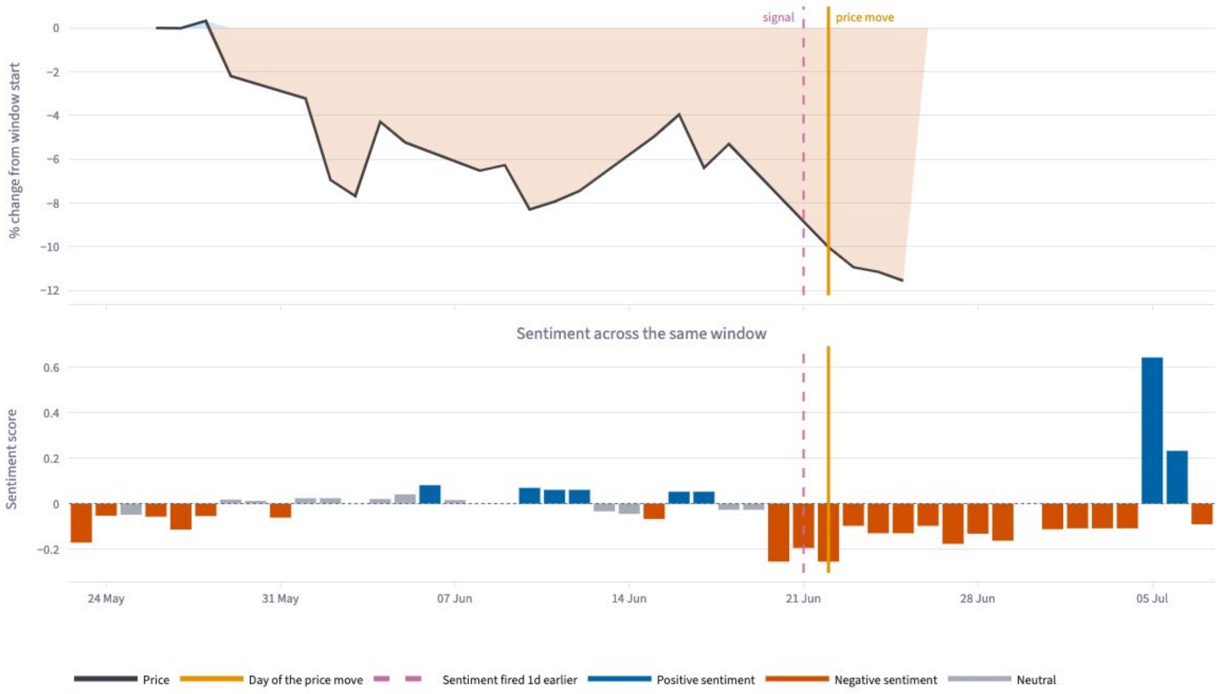


Key Moments

Zooming in on days when a stock moved abnormally — beyond two standard deviations — to see what the chatter looked like in the weeks beforehand.

Did chatter move first? Sentiment fired 1 day before GOOGL moved -5.0%

22 June 2026 · signal strength -0.195 · sources: 0

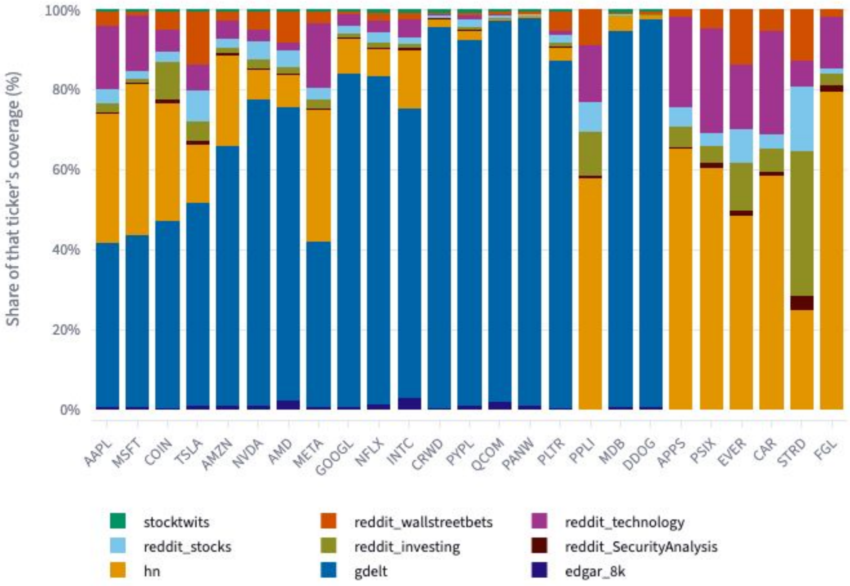


Sources

Six independent feeds contribute to every sentiment reading. They differ enormously in how much they publish, and separately in how strongly they lean.

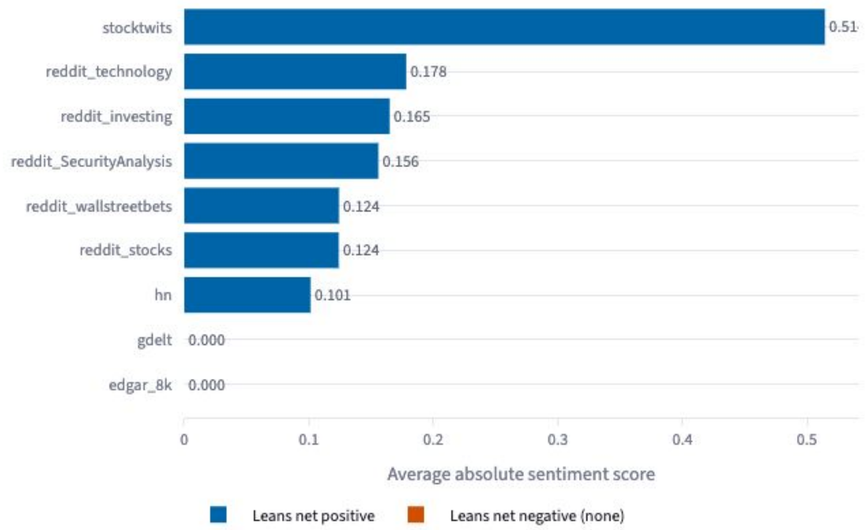
Which source supplies the most coverage? gdelt

Each bar is one ticker's coverage split by source, ordered by total volume. Showing 25 tickers



Which source words things most strongly? stocktwits

Bar length is how forcefully a source phrases things, not how often it is right



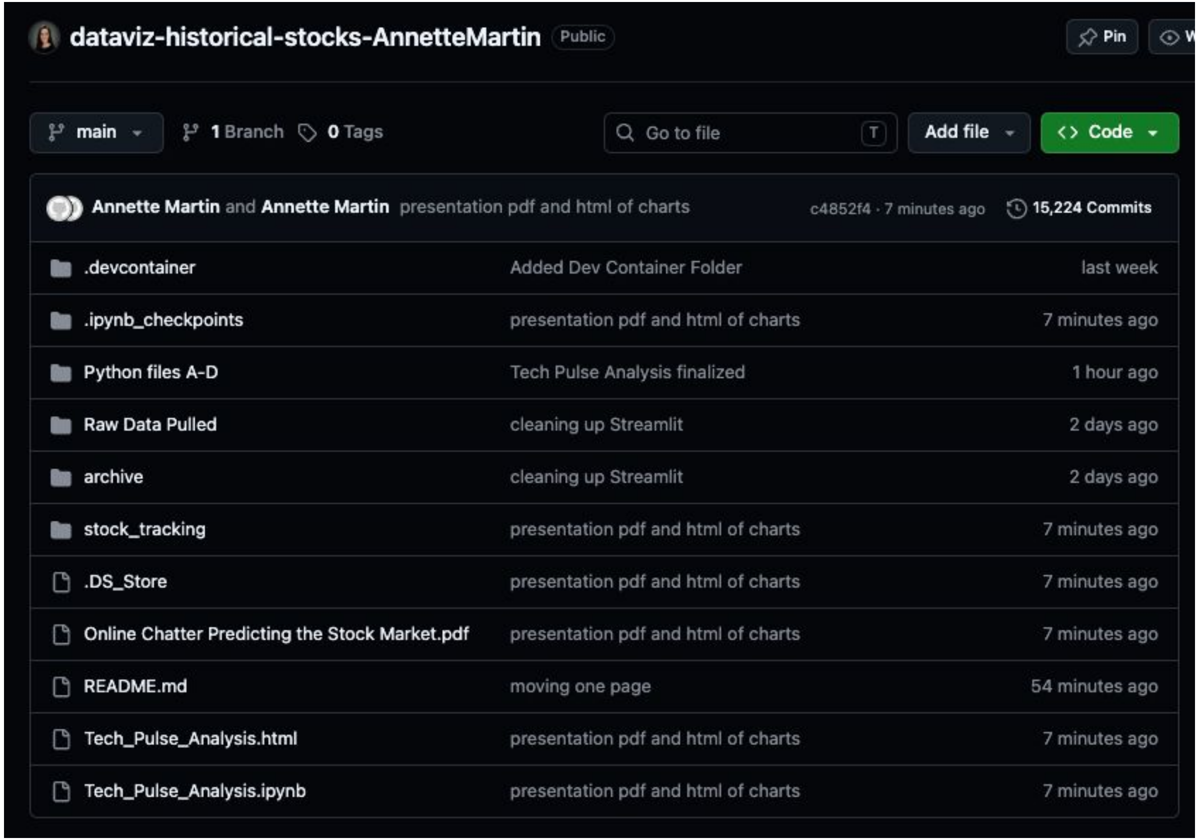
Strength of language and accuracy of prediction are separate properties. A source can be consistently emphatic and consistently wrong.

Future work

Swap VADER for FinBERT. VADER matches words against a general-English lexicon and knows nothing about finance — "beat expectations" scores positive because *beat* is a positive word, while "missed estimates" barely registers. FinBERT is fine-tuned on analyst reports and financial news, and reads context instead.

That matters most for the unresolved negative-signal result above. If VADER is systematically misreading bad news, the negative sample is contaminated rather than merely small, and FinBERT would settle whether the below-chance reading is real. Scoring full articles rather than headlines, and using FinBERT's confidence scores to drop uncertain readings, would both help. The cost is compute: VADER runs locally in minutes, FinBERT needs GPU inference across eleven years of six sources.

Also worth pursuing: intraday resolution, explicit market-regime conditioning given the post-2021 decay, and testing markets outside the US.



Github:
https://github.com/annhmartin/dataviz-historical-stocks-AnnetteMartin/tree/main
Streamlit:
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